PARMCO - Checkpoint 1

AI-generated assembly instructions and actual reported parts list | Updated October 19, 2026

Current status: The motor stand and both bypass capacitors are installed. The four SN754410NE input pulldown resistors are 1.8 kohm each. The lab partner reports the motor worked during physical testing. Electrical stepping signals for both commanded directions were verified with a multimeter; actual physical direction and exact one-revolution behavior are not independently documented.

Parts list

Component	Qty	Use / status
Raspberry Pi 4 Model B Rev 1.2	1	Controller; BCM GPIO
SN754410NE DIP-16	1	Dual H-bridge
4N25 DIP-6	1	Optocoupled enable
NEMA17 XY42STH34-0354A	1	Motor; tested per partner
External 5 V and 12 V bench supply	1	Logic and motor rails
330 ohm resistor (R1)	1	GPIO24 to 4N25 LED
10 kohm resistor (R2)	1	Enable net pulldown
1.8 kohm resistors (R3-R6)	4	SN input pulldowns; installed
100 nF capacitors (C1-C2)	2	Supply bypass; installed
Breadboard and jumper wires	As used	Interconnections
Motor stand	1	Installed
Propeller	As used	Stall-speed test; status confirm
Digital multimeter, safety eyewear	As used	Verification and safety

Assembly instructions

1. With all supplies off and RP4 unplugged, orient the ICs using their package notches and verify pin numbering against the datasheets.

2. Wire SN pin 16 to external 5 V (VCC1) and pin 8 to external 12 V (VCC2). Connect SN pins 4, 5, 12, 13, RP4 GND and bench-supply ground together.

3. Wire RP4 GPIO24 (physical 18) through R1 (330 ohm) to 4N25 pin 1; 4N25 pin 2 to GND; pin 5 to external 5 V; pin 4 to SN pins 1 and 9. Install R2 (10 kohm) from this enable node to GND. Leave 4N25 pins 3 and 6 unconnected.

4. Connect RP4 GPIO17, GPIO27, GPIO22 and GPIO23 to SN input pins 2, 7, 10 and 15, respectively. Install one 1.8 kohm pulldown from EACH input pin to common ground (R3-R6).

5. Install C1 (100 nF) from SN VCC1 pin 16 to GND and C2 (100 nF) from SN VCC2 pin 8 to GND; both capacitors were reported installed.

6. With power off, confirm winding pairs by resistance. Green/grey (~37 ohm) connect to SN outputs pins 3/6; red/yellow (~36.3 ohm) connect to outputs 11/14. Secure motor on installed stand before powered operation.

7. Complete and record the student-written pre-power inspection and use instructor-directed current limits before motor-powered operation. SN754410NE does not regulate winding current.

Verification and motor testing

Verified by measurement: Full-step clockwise and counterclockwise commanded GPIO sequences matched the expected HIGH (~3.14-3.17 V) and LOW (~0 V) patterns at SN input pins 2, 7, 10, and 15. ARM generic timer frequency was read as 54,000,000 Hz. The 4N25 enable signal reached approximately 5 V HIGH.

Reported by lab partner: Motor was physically tested and worked. Do not claim physical direction, exact revolution count, or stall speed as verified until the partner provides the observation or measured data.

Stall-speed entry: Record each measured last successful steps/second and convert to RPM = steps/sec x 60 / 200 in full-step mode. Include repeated trials and comparison with the 200 RPM project target.

Safety and submission notes

Disconnect power before moving wires; keep the motor secured; wear eye protection whenever a propeller spins. The pre-power inspection is a separate **student-written** requirement and must reflect real inspection observations. This AI-generated document should be checked against the final physical build before ELC submission.